

Post-Brachytherapy Pigment Dispersion in Uveal Melanoma

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Dear Editor,

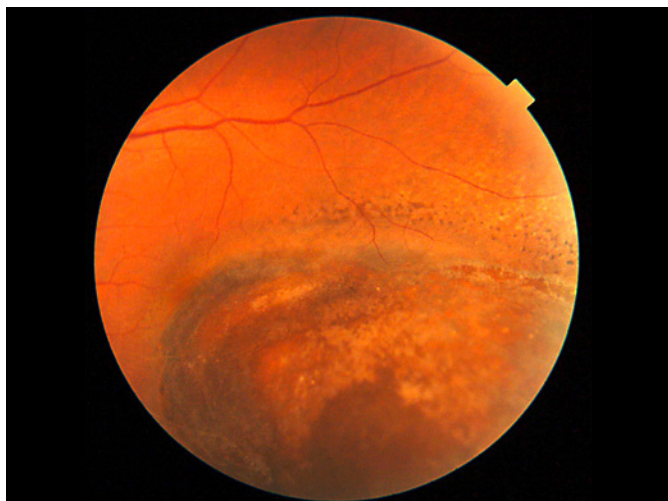
We read with great interest the article titled “Post-Brachytherapy Pigment Dispersion in Uveal Melanoma” in a recent edition [1]. The article throws light on an unusual but known complication of brachytherapy in the form of delayed intravitreal pigment dispersion post-iodine-125 plaque brachytherapy in 4 patients with choroidal melanoma with no tumor recurrence. We would like to share similar experience from our institute, the only major difference being the use of ruthenium-106 plaque brachytherapy in our patients.

The first patient is a 46-year-old lady, diagnosed with a fovea-involving choroidal melanoma measuring $14.1 \times 12 \times 5.94$ mm in 2016. She underwent brachytherapy with a notched Ru-106 plaque for a period of 12 days, followed by trans-pupillary thermotherapy. The lesion started regressing a month post-brachytherapy. She received 4 more sessions of TTT until 2017 and was under regular follow-up. Seventeen months after brachytherapy, at a regular follow-up visit, she was found to have dispersed pigments in the vitreous cavity (Fig. 1, 2). The tumor was regressing and did not show any signs of recurrence; hence, she was followed up closely. Even after 4 years, the pigments remain unchanged with no evi-

dence of tumor recurrence or metastasis. At the last visit, the tumor measured $12.75 \times 10.97 \times 4.5$ mm, and there was no lateral or anteroposterior increase in size or extent of the tumor.

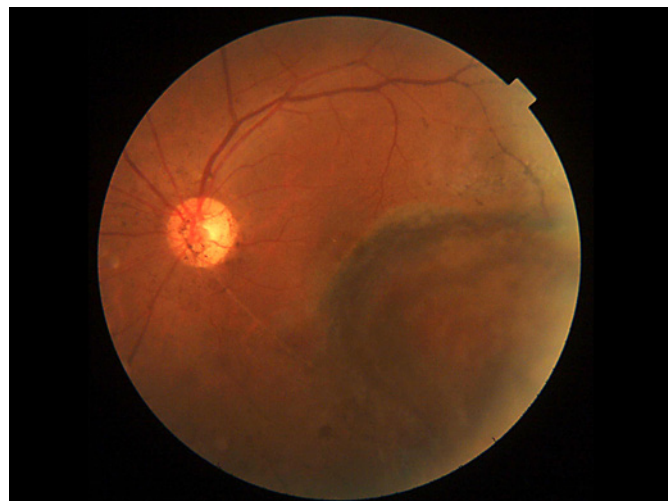
Our second patient is a 34-year-old gentleman who presented to us first in 2015 with a peripapillary choroidal melanoma measuring $6.9 \times 5.7 \times 2$ mm abutting the disc. He underwent brachytherapy with Ru-106 plaque for 2 days. His tumor started regressing 3 months post-brachytherapy. Twenty-two months later, on a regular follow-up visit, he was found to have pigment dispersion in vitreous cavity with no recurrence of the tumor. He has been under close follow-up with persisting pigments and no circumferential growth or increase in height of the tumor or metastasis.

Our last patient is a 46-year-old gentleman who presented to us in November 2020 with a typical mushroom shaped choroidal melanoma of dimensions $13.87 \times 12.2 \times 7.04$ mm nasal to the disc with inferior exudative retinal detachment. He underwent brachytherapy with Ru-106 plaque for 2 days and TTT later as adjuvant therapy. The tumor started regressing 3 months post-brachytherapy, and the patient was under regular follow-up. Twelve months after brachytherapy, the patient developed pig-



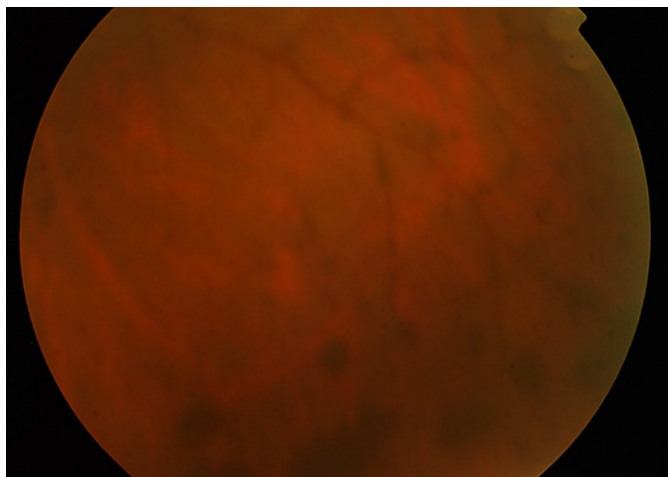
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Fig. 1. Pigment dispersion seen in the vitreous cavity and above the mass lesion in patient 1.



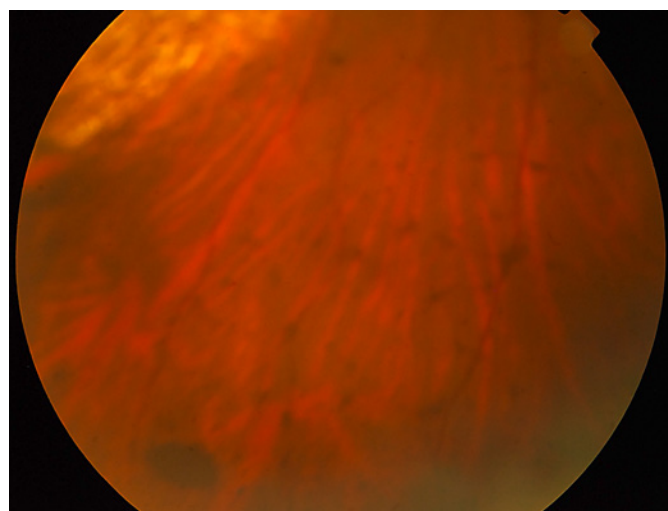
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Fig. 2. Pigment dispersion over optic disc in patient 1.



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Fig. 3. Pigment dispersion seen in the vitreous cavity inferiorly in patient 3.



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Fig. 4. Pigment dispersion seen in the vitreous cavity inferiorly in patient 3.

ment dispersion in vitreous cavity with no evidence of tumor recurrence or metastasis (Fig. 3, 4). He is under regular follow-up, and at the last visit, the tumor measured $9.41 \times 7.54 \times 5.15$ mm.

This rare, but interesting entity of delayed pigment dispersion post-brachytherapy in melanoma, as quoted by the authors, could be due to brachytherapy-induced tumor shrinkage causing pigment release or secondary to TTT induced ischemic necrosis and can mimic tumor recurrence. However, with the tumor being stable in terms of no increase in dimensions, no extraocular extension,

no intraocular seeding, and no evidence of metastasis, it can be safely observed with close follow-up. Our patients have been closely followed up and displayed no evidence of tumor recurrence.

Statement of Ethics

Written informed consent was obtained from the patient for publication of the details of their medical case and any accompanying images.

Conflict of Interest Statement

The authors have no conflicts of interest to declare.

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Author Contributions

Dr. Mahesh Palanivelu Shanmugam – conception of work and final approval of the version to be completed. Dr. Shwetha Suryakanth – drafting the work. Dr. Rajesh Ramanjulu – final approval of the version to be completed. Dr. Divyansh Kailash Chandra Mishra – final approval of the version to be completed.

Reference

- 1 Maniar A, Marr BP. Post-brachytherapy pigment dispersion in uveal melanoma. [Ocul Oncol Pathol](#). 2021;7(6):396–400.